

Wireless Sensor Networks in Industry 4.0: WirelessHART and ISA100.11a

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Abstract—Industrial Internet of Things (IIoT) is a key technology in the scenario of Industry 4.0. Just like any IoT architecture, Wireless Sensor Network (WSN) plays a fundamental role, but the factory floor presents challenges that make usual WSN technologies inadequate. This paper addresses and compares the main features of the two WSN standards developed specially for an industrial application, WirelessHART and ISA100.11a. The main contribution of this work is the comparison between them in the context of IIoT in Industry 4.0. The results show that the two standards have many similarities and almost equal performances in factory floor conditions, and can even coexist. However, ISA100 has some features that make it more aligned with the Industry 4.0 concepts.

Index Terms— Industrial IIoT; Industry 4.0; Fieldbus; WirelessHART; ISA100.11a

I. INTRODUCTION

Industry 4.0 (I4.0) is a concept that could be defined as the integration in manufacturing process of technologies, such as Big Data Analytics, Cloud Services, 3D-Printing, Cyber Security, Autonomous Robots, Internet of Things (IoT), Augmented Reality (AR), Simulation, Horizontal and Vertical Integration [1].

It is necessary to differentiate the concepts of Industrial IoT (IIoT) and industrial networks (called *fieldbuses* and *device-buses*), although they overlap. Although industrial networks have existed for last decades, and more recently some of them make use of Ethernet and Internet technologies, Industrial IoT (IIoT) is a more comprehensive concept. To face I4.0 challenges (higher degree of interconnection, cognitive automation, shifting information collection, and processing into cloud-based applications [3]), IIoT goes far beyond the use of machine communication in control process. Whereas some authors understand IoT simply as a global network of smart objects that are able to share information between itself [4] [5], other authors see IoT as a system architecture. In [6] IoT is defined as “interconnection of sensing and actuating devices providing the ability to share information across platforms through a unified framework, developing a common operating picture for enabling innovative applications”. There are also authors who argue that an IoT solution should have the addressing of their devices made through IP addresses. [23]. In the industrial context, this definition means that the sensor and actuator data can flow and be useful in many sectors of manufacture process and management. Moreover, it makes clear the need to collect data not only from process variables (for control purposes) but also from a diversity of production process data that could be used in management, traceability, maintenance, logistics, etc.

An IIoT system is composed of software and hardware, where the hardware is composed of smart sensors and actuator and its network infrastructure; this sensor network typically uses wireless technology, so it is a Wireless Sensor Network (WSN) and is the focus of this work. Because there are specific requirements stemming from the nature and environment of industrial applications, as discussed in [7] [8], industrial networks have specific features, even if they share aspects or characteristics of networks from other markets. Notwithstanding some authors considered that IIoT can use IEEE 802.11 or LTE in WSN [6] [3], in the practices the most used industrial wireless networks protocols, WirelessHART (WH) and ISA100.11a (ISA100). The main contributions of its paper is present and compare the characteristics of these network standards as WSN in I4.0 context.

The remainder of this article is organized as follows: since both WH and ISA100 use the IEEE802.15.4 radio standard as the physical layer and MAC sublayer technology, Section II presents a review of this standard, where special attention is given to time-slotted channel hopping (TSCH). Section III and Section IV describes the upper layers of WH and ISA100, respectively. Section V shows a comparison between these standards and the Section VI present the conclusions.

II. THE STANDARD IEEE 802.15.4

The IEEE 802.15.4 standard specifies medium access sublayer (MAC) and physical layer (PHY) characteristics. Layers one and two of OSI/ISO model defines various features of a technology, such as network topology, power control mechanisms, packet detection, and retransmission techniques. IEEE 802.15.4 addresses Low-Rate Wireless Personal Area Network (LR-WPAN) in low cost, low power consumption and low data rate applications. Its characteristics make it to be used in several communication technologies in IIoT such as Zigbee, Thread, and, of special interest in this study, WH and ISA100 [10]. The standard was developed for low-power concerns, since many applications in mind are battery-powered and constant exchange of these is undesirable, as is the case in industrial applications. For a longer battery life it is necessary that the application establish a duty cycle with operating time and reduced power state (sleep mode) [11]. There is a compromise between battery life and tolerable latency. This relationship is especially critical in industrial applications since the sampling rate of the control process variables is a function of the response time of this process.

a) *Radio characteristics:* WH and ISA100 technologies operate in the 2.4GHz range (free ISM band worldwide)

using O-QPSK modulation with DSSS (Direct Sequence Spread Spectrum). DSSS is used to increase immunity and coexistence with other ISM band transmissions (such as IEEE 802.11) and not for multiple medium access purposes [14]. The transmission power of each device for WH and ISA100 must be adjustable between +10dBm and -10dBm. There is a commitment relationship between transmission power, distance and battery life. Devices must have a minimum sensitivity of -85dBm or better [15]. These means in an outdoor free line of sight condition, with isotropic antenna, the distance between nodes could be greater than 500m; in industrial indoor environment, these ranges can fall drastically. There are sixteen adjacent channels with 5MHz bandwidth and center frequencies ranging from 2.405 to 2.480 GHz. Channel sixteenth is not normalized in some regions of world, so WH do not uses them and for ISA100 it is optional. As can be observed, this band overlaps with one of the IEEE802.11 bands.

b) Other PHY functions: In addition to providing the modulation and transmission, others function must be performed by the physical layer. The Energy Detection (ED) function informs the power level in the channel. For each received frame, the radio updates a Link Quality Indicator (LQI). There is also a free channel indicator (CCA) function used by the MAC layer for CSMA implementation.

c) Devices roles: A LR-WPAN consists of two or more devices in communication. There are two types of devices: Reduced Function Device (RFD) and Full Function Device (FFD). These devices can take on the roles of PAN Coordinator, Coordinator, or a simple Node of the network. Only FFDs can assume roles as Coordinator or PAN Coordinator, since these functions require more processing and energy resources. In the WH and ISA100 technologies, all elements of the network have FFD capability, although the routing function can be disabled in ISA100, making device a simple terminal node.

d) Data Link Layer Addressing: In an IEEE 802.15.4 network, a FFD must play the role of PAN coordinator, which creates the network, assigns addresses and controls the devices. Each network has a unique identifier (PAN ID). Every device on the network has a long address and can use a short address. The long 64-bit addresses are of the extended universal identifier type (EUI-64) assigned by IEEE Registration Authority. The 16-bit short addresses works like nicknames of local validity assigned by the PAN coordinator.

e) Data Link Layer Topologies: An IEEE 802.15.4 network can assume two topologies at the link-layer level: star and peer-to-peer [12], as can be seen in Fig. 1. In star topology, the communication happens between each device and PAN coordinator only. In the peer-to-peer, topology there is also the PAN coordinator, with the same functions, but devices can communicate with each other (as long as they are in their radio range). The peer-to-peer topology allows routing modes where the message can "hop" from device to another until it reaches its destination, allowing mesh networks. The definition of mesh or multi-hop routing topology, as well as routing strategies, are defined in upper layers

f) Time-Slotted Channel Hopping: The mode of medium access used by WH and ISA100 is Time Slot-

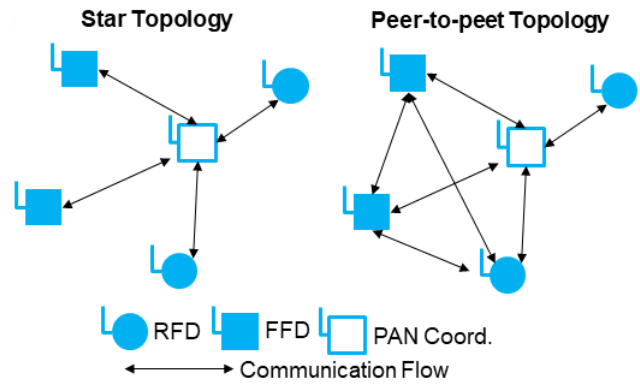


Fig. 1. Illustration of the supported topologies by the IEEE 802.15.4.

ted Channel Hopping (TSCH), added to revision E of IEEE802.15.4 in 2008. TSCH is used together as CSMA-CA and increases the robustness and security in the network, besides guaranteeing a deterministic behaviour. This strategy of access to medium consists in setting up a scheduling table that correlates instants of time and channels (Fig. 2). Each peer communication in the network takes place in a predefined time slot, that is, a Time Division Multiple Access (TDMA) scheme. The set of time slots constitutes a *slotframe*, periodically repeated. The duration of the time interval can be fixed (as in WH) or adjustable (as in ISA100). Within

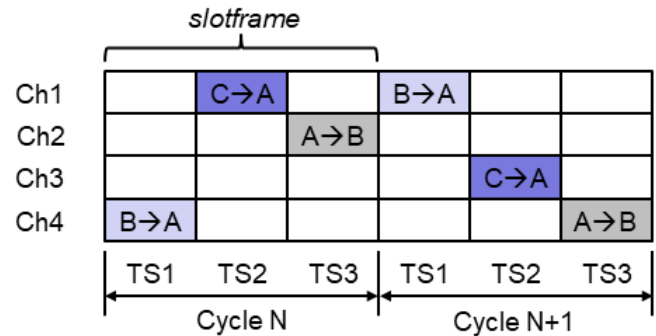


Fig. 2. Example of time-slotted channel hopping scheduling table.

a time slot, it is possible to establish a single communication exchange between two devices composed of a transmission and an acknowledgment. Before each transmission, however, a channel listening is made to ensure that there are no other transmissions, through the CCA function. Fig. 3 shows the time diagram of this process.

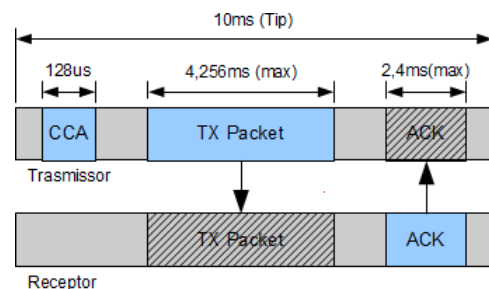


Fig. 3. Packet transmission and acknowledgment timing.

TSCH also uses channel hopping: at each time slot, the communication occurs in a different available channel. WH and ISA100 perform channel hop according to pseudo-random sequences, so that the same communication does not always happen on the same channel. In this way, persistent interference on one channel (for example, caused by other ISM band communications) will not always affect one communication or the entire system. This functionality also increases security, since it is necessary for any invading element to know the sequence of jumps to be able to follow the exchange of information in the system.

The combination of frequency hopping with TDMA establishes a system of diversity in time and frequency. In this system, frequency offset and time offset defines a "link", for a unique communication between two devices. IEEE 802.15.4 establishes how the scheduling table is executed, but upper layers of the network define the table assembly and link assignment and WH and ISA100 adopt different scheduling schemes.

g) *ARQ and Blacklisting*: WH and ISA100 implement the ARQ (Automatic Repeat reQuest), which consists of automatically retransmitting every message that was not acknowledged, ensuring that, in combination with TSCH, the message is retransmitted in the next *slotframe* on a different channel. These protocols also associate TSCH with the blacklisting functionality, which is to create a list of channels to avoid because they have been identified with the presence of interference. This functionality allows for increased reliability and is implemented using channel occupancy indicators and communication quality (ED, CCA and LQI).

h) Security: As a security function for the MAC layer, the IEEE 802.15.4 standard features 128-bit AES encryption algorithms. The purpose of this encryption is to ensure data confidentiality and authenticity, and detection of duplicate information. The IEEE standard does not define how the security key management should be or which authentication policy should be applied, these settings come from higher layers of WH and ISA100.

III. WIRELESSHART / IEC 62591

The WH protocol was developed from HART (Highway Addressable Remote Transducer) industrial wired protocol, aiming at compatibility with field instruments, applications and tools already installed [16]. It was proposed by the HART Communication Foundation (now FieldComm Group) and was established as an international standard (IEC 62591-1) in March 2010. A WH field wireless network is typically connected to an industrial controller (PLC ou DCS) for control purpose by a fieldbus (MODBUS, Ethernet/IP, etc). Other IIoT application accesses data from WH sensors and actuators through some proxy, e.g., an OPC server. Fig. 4 shows an IIoT system architecture using WH.

a) *Application Layer*: The application layer protocol used is HART 7 [17], which ensures compatibility with other HART applications already implemented [18]. At application layer, WH is command-oriented. Application layer define commands, responses, data types and status of devices. Security key management is also at the application layer [19].

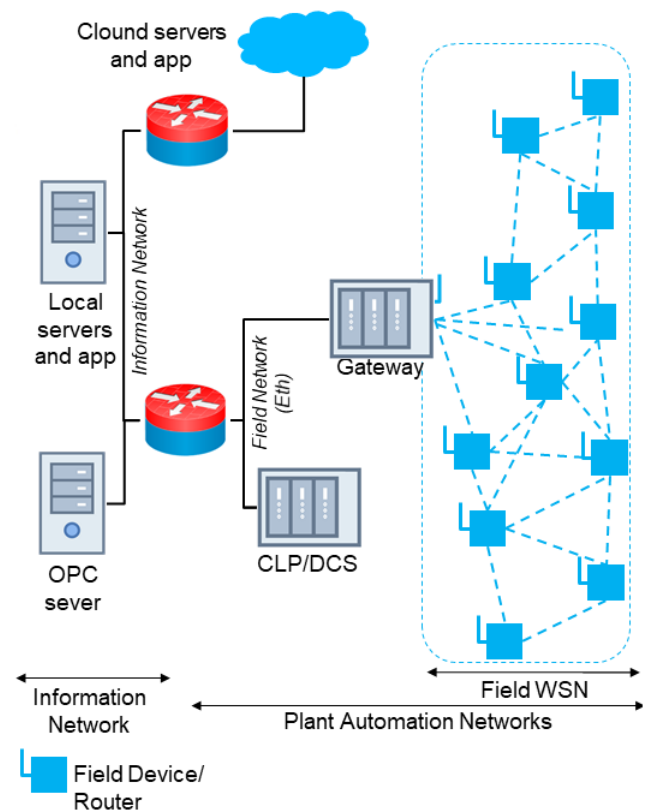


Fig. 4. WH network architecture for industry 4.0 application.

b) Transport Layer: The transport layer provides reliable, non-connection-oriented communication. Optionally end-to-end acknowledgments can be used in the network.

c) *Network Layer*: At the network layer, the WH provides a full mesh topology. For this, every network device has router function and the ability to find neighbours nodes and measure the radio frequency (RF) signal strengths to establish paths and links with other instruments. WH supports two routing schemes: source routing and graph routing. In source routing, a single routing path is indicated in the packet header; each element in this path forwards the packet to the next. There are no alternative paths or redundancy. In graphical routing, the indication in the packet header indicates paths that can contain multiple paths that collaborate so that at least one copy of the packet reaches the destination. Three types of graphs are defined: uplink, downlink and broadcast. In both cases, the paths are defined and maintained by the network manager [20]. The addressing within the network is done by a 16-bit WH address that allows more than 30,000 elements in a single network. The network is composed of devices such as:

- Field devices: sensors and field actuators. All field devices must be able to route packets.
- Routers: router devices (without sensor functions) can be installed for extension or redundancy purposes;
- Adapters: for connecting instruments with HART wire interface;
- Hand-held: for service and maintenance staff;
- Gateway (single or redundant): acts as a bridge to the control application. Typically, it also aggregates the

functions of Network Manager, Security Manager and Access Point.

d) *Data Link Layer*: In the link layer is used the TSCH with fixed time slot of 10 ms. Channel hopping follows a pseudo-random sequence. The schedule table is set up to guarantee each instrument the programmed sampling rate. The blacklisting system is manual [21] and should be done by the network administrator according to the field conditions.

IV. ISA100.10A / IEC 62734

ISA100.11a was developed by International Society of Automation (ISA) [3] [9] and became an ANSI standard in 2011 and later an international standard IEC 62734. It is part of a family of standards designed to support a wide range of wireless industrial plant needs including process automation, factory automation and RFID, called ISA100. The main design criteria for ISA100 include flexibility, support for multiple applications, high reliability and integrated security. As showed in Fig. 5, a ISA100 network is actually composed by a field WSN and a wired backbone. Although a subnet may have many devices, two subnet cannot communicate by radio interface, it is necessary to rout packets through access points and backbone (cause this, access points is more properly called backbone routers). The backbone extend the ISA100 network to cover a large plant and can use segments of communication technologies like fiber optics or radio links. Between backbone and the fieldbus, there is a Field Wireless Manager (FWM) station that acts as a gateway and entire network management. This station communicate with control system (PLC or DCS) through a specified fieldbus protocol and can aggregate an OPC server too.

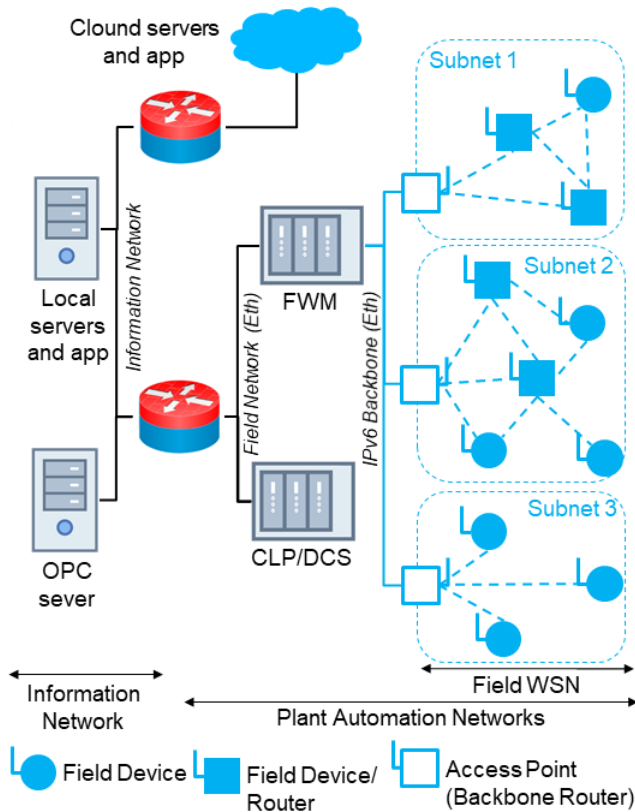


Fig. 5. ISA100 network architecture for industry 4.0 application.

a) *Application Layer*: ISA100 is based on open standards such as IPv6 and UDP. It does not specify an application layer protocol or an interface to an existing protocol; this is a choice of each gateway manufacturer to make available one or more fieldbus protocols. Application layer define software objects to model physical devices. The application layer is also responsible for managing the security keys. ISA100.11a is suitable for control applications (closed loop control, closed loop supervision and open loop control) and monitoring (alert, logging and download and upload) and is not recommended for security applications.

b) *Transport Layer*: In the transport layer UDP (User Datagram Protocol), which is an unreliable, non-connection-oriented protocol. UDP is an often used in industrial networks due to higher speed and lower overhead when compared with TCP, getting the reliability of information exchange guaranteed by link layer protocols.

c) *Network Layer*: The ISA100 network is divided into a wired segment (backbone) and wireless subnets. The backbone connects the Field Wireless Manager (FWM) to multiple Access Points (APs). In this segment of the network, IPv6 routing is properly used. The wireless segment consists of subnets that interconnect through the backbone. IPv6 is used for addressing, guaranteeing a *de facto* IP address for each element of the network. 6LowPAN adapts the IPV6 for IEEE 802.15.4 operation. However, routing at the network layer level happens on the backbone, that is, between APs and FWM. Within the subnetworks, routing happens at layer 2, as will be discussed. Packet segmentation and reassembly occurs in the APs. The network is composed of devices such as:

- Field Wireless Manager (FWM): manages network and security and acts as a gateway for other networks.
- Access Point (AP) or backbone router: it has the function of connecting backbone with the wireless elements.
- Media converter (MC): physical media adapters used for network extension expansion (e.g. fiber optic adapter).
- Integrated Gateway (IG): for less complex networks, this element aggregates the functions of FWM and AP; in networks with IG there is no backbone.
- Field wireless devices (FWD): these are the field devices.

d) *Data Link Layer*: At the data link layer TSCH is used, but with slot time configurable from 10ms to 14ms. Frequency hopping follow a pseudo-random sequence and ARQ usage ensures retransmission on different channels. There is also the option for fast frequency jumps (the channel changes at each interval), slow (the channel changes at each *slotframe*) and mixed (combination of the previous ones). The blacklisting process is automatically implemented. Within each subnet, the routing does not effectively use IPv6, but the short 16-bit address allocated locally. In fact, routing on a subnet occurs at the link layer rather than the network layer counteracting the division of layer functions of ISO/OSI model. Within the subnets, routing can be by graph or by source, as in WH [22]. To ensure lower battery consumption, it is possible to disable the router function of a device. In practice, therefore, the possible topologies are mesh, star, and mesh-star [18]. Special care must be taken in configuring

devices as a router or not to ensure coverage and reliability (redundancy) of the network.

V. DISCUSSION AND ANALYSIS

Despite the similarities, the standards have many differences. An extensive discussion of similarities and differences is made in [17], [18] and [24]. Table I shows the most relevant differences. Both networks present robustness (TSCH, redundancy topologies) and security (cryptography, channel hopping) features that make them reliable in industrial applications. In addition, low latency and deterministic temporal behaviour make them suitable for industrial control applications.

TABLE I
COMPARISON BETWEEN WH AND ISA100

WirelessHART	ISA100.11a
Every field instrument must be a router (real mesh network)	Field instrument may not be a router (battery safe).
Routing at the network layer; 16-bit address	Wireless subnet: Routing at DLL using subnet short address; Backbone: routing at network layer using IPv6
Time slot of 10ms, a unique channel hopping algorithm	Configurable time slot, 3 types of channel hopping, 5 hop sequences (may cause interoperability problems)
Complete stack network in wireless technology, but it is not IP based	It is an IPv6 network in the backbone connecting wireless subnets
Supports unicast and broadcast	Supports unicast, broadcast, duocast, and n-cast.
Transport layer with acknowledgement (more reliable).	UDP-based transport layer (faster).
Application layer uses HART 7	Does not define application protocol
Command oriented: command-response protocol	Object Oriented: virtual modeling of devices

WH allows an extensive mesh network and can have multiple AP, i.e., many connection points of the wireless network with the wired infrastructure. In ISA100, backbone routers concentrates the flow from different subnets, but many subnets connected by a backbone can create a large wireless network. Against the common sense, the creation of large mesh networks can become a problem, since as the number of devices increases, consequently increases the traffic in the network, and some nodes routers can eventually be over-demanded, thus causing network fragility and excessive energy consumption.

In ISA100, since addressing uses IPv6, there is no practical address limitation. A single WH network is limited by address space to about 30,000 devices. Despite the address space, in both networks in practice each AP is able to coordinate a maximum of 50 to 100 devices [18]. Other differences is that ISA100 performs fragmentation and reassembly at the network layer while WH does it at the application layer. WH support unicast and broadcast schemes, while ISA100 further supports duo-cast, and n-cast (n-cast differs from multi-cast because all receiver acknowledges received packets).

For IIoT context there is an important difference between WH and ISA100: While WH specifies a set of commands ISA100 define devices models to represent virtually physical objects. That means while WH is command-oriented, ISA100

is object-oriented. Object-oriented has better compatibility between different manufactures and is more suitable for IIoT virtual twin representation [24].

Coexistence between WH and ISA100 and between both and IEEE 802.11 is a very important performance question. An experimental comparison of packet loss rate (PLR) in [25] shows that interference between the two industrial WSN is very small (increase almost 0.5% PLR), even when compared with PLR in industrial environment without interference (1% for both.) In turn, the effect of an IEEE802.11 interfering network was higher, causing an increase of less than 2.5% in both. In the presence of this type of interference, the performance of the networks is almost identical. Considering co-existence of the whole network, [26] makes a case study of the implementation of ISA100 and WH networks for temperature monitoring in the same industrial environment. As a result, the WSN coexist in the field, the gateways belonged to the same network and the data are integrated in the same historian database.

ISA100 has configurable time intervals (10 to 14ms) while WH specifies only a 10 ms interval. The channel hopping mechanism supports multiple configurations on the ISA100 since WH specifies only one channel hopping mechanism. As for the wireless network topology, all ISA100 devices do not must to support routing while WH devices must do so. This implies that WH is indeed a mesh network whereas ISA allows mesh, star and star-mesh implementations, and impacts in power consumption of devices.

About security, in WH all security features are mandatory while in ISA100 they are optional. Since security schemes consume additional time, memory, and power, WH devices cannot deactivate them for benefits such as longer battery life. In contrast, the option to disable security features can make the network susceptible to attack.

From the standpoint of implementation, WH is relatively simple, with very few configuration options or parameters. ISA100, on the other hand, is a complex specification with many configurations and optional parameters. The variety of options may be a difficulty when equipment from different manufacturers is installed to operate on the same network [21].

Regarding the characteristics relevant to the industry context 4.0, it should be noted that the main aspect is interoperability with other applications in the plant management network. In this criterion both networks would be accessed via some proxy system, of which the most common would be an OPC/UA server. However, the fact that the ISA100 network uses IP address on all its devices allows (at least potentially) that these are accessed directly by the applications on the corporate network. Considering that the use of wireless networks allows an increased coverage of devices in the industrial area, the two standards have different coverage strategies. By using mesh network with routing on all devices, WH allows for increased coverage without reliance on additional infrastructure at the cost of being exposed to overload in some nodes. On the other hand, the expansion of ISA100 coverage is done by extending the backbone and adding more APs; although the backbone uses Ethernet technology, therefore cheap and easy to access, it can become a weak

point of the system, being important to evaluate strategies of redundancies in critical segments.

VI. CONCLUSION

The perspective of industry 4.0 allows predicting an increase in the number of sensors, seeking a greater connectivity and volume of information of the productive process, consequently leading to the growth of networks in industrial environments. Wireless networks have great advantages in terms of cost of installation, maintenance, flexibility and modularity. The challenges of the factory floor environment require adequate networks and reliability, challenges coupled with the need for vertical integration over the Internet and interoperability via open standards.

IEEE 802.15.4 is unquestionably the radio standard for WSN and TSCH allows robustness and immunity to interference to the factory floor. WH and ISA100 have many similarities, their main differences are in network organization and in the application layer. Only ISA100 is an IP network (IPv6); according to some definitions for IoT only it could indeed be considered IoT network. Additionally, ISA100 has more flexible configuration and diverse system options. However, numerous articles point to the simplicity (or fewer options) of WH as an implementation and interoperability advantage that makes it more widely used in practice.

For the goal of a smart factory, is important that the WSN technology must be trustworthy. The papers analysed and features compared shows that WH and ISA100 performs similarly in this criterion. Nevertheless, ISA100 has features that make it more aligned with I4.0 concepts, which are (i) it is an IPv6 network that supports typical Internet open protocols, (ii) application layer is object-oriented and (iii) is more flexible in many aspects (cast schemes, timing, device's roles, security).

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